
P M - 0 4 - P S 0 2

Automation Design & Implementation

Robotic Process Automation Solution

ASSESSMENT EVIDENCE PACKAGE

<https://github.com/TheFish1011/PM-04-PS02.git>

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1. Purpose of this document

This document provides the design, implementation, and execution evidence for the MzansiConnect Customer Onboarding automation. The solution automates the onboarding of customer records from a CSV file into a CRM system, including validation, logging, and exception handling.

This document maps to the following assessment criteria:

- PA0201 – Decide Bot Steps
- PA0202 – Logging Behaviour
- PA0203 – Exception Handling
- IAC0201 – Reusability
- IAC0202 – Logging & PII Handling
- IAC0203 – Exception Handling Validation

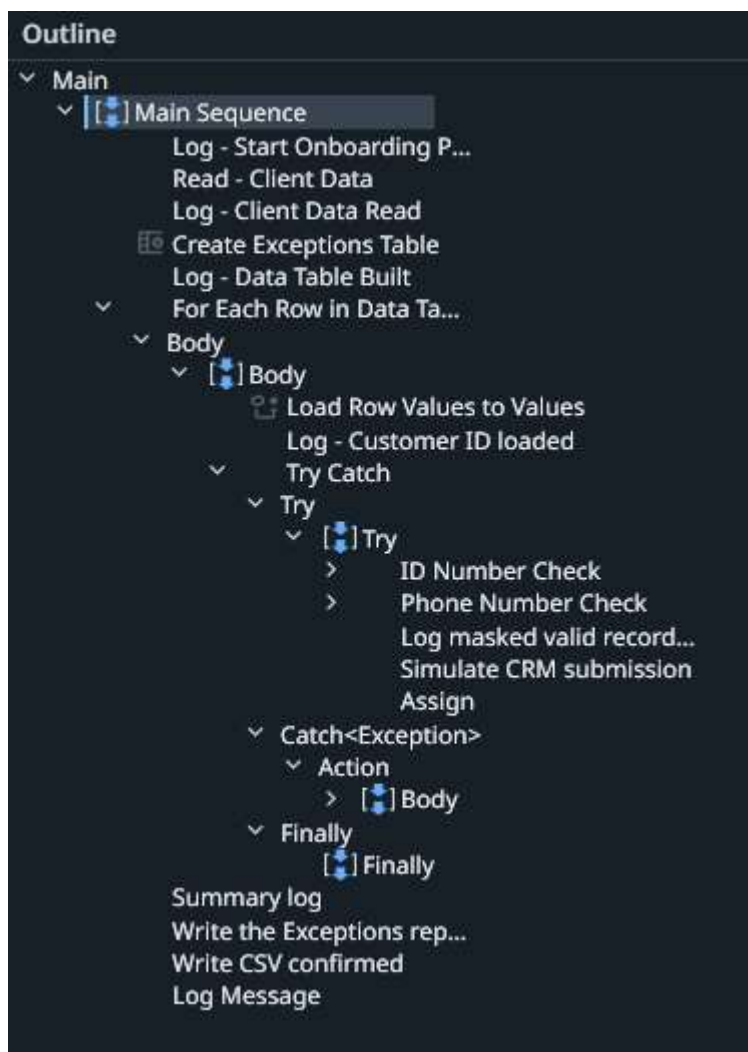
2. Process Analysis and Bot Steps (PA0201)

2.1 Overview

1. Start process and log execution start time
2. Read customer data from CSV file
3. Create exceptions tracking table
4. Loop through each customer record
5. Validate data fields (SA ID and Phone Number)
6. Process valid records into CRM
7. Log outcomes for each record
8. Generate summary and exception report
9. End process

2.2 Detailed Sequential Steps

1. Open CSV file containing customer data
2. Load data into DataTable
3. Open CRM system (simulated in this implementation)
4. For each customer record:
 1. Extract Name, Surname, SA ID, Phone Number
 2. Validate SA ID presence
 3. Validate Phone Number format
 4. If valid:
 1. Submit data to CRM
 2. Log success
 5. If invalid:
 1. Log failure
 2. Store in exception report
6. Generate summary report
7. Save exception report to CSV
8. Log process completion



3. Logging Behaviour (PA0202 & IAC0202)

3.1 Information Logged

- Process start and end timestamps
- Total records processed
- Per-record processing status
- Validation failures
- System exceptions
- Final summary (Success / Skipped / Total)

3.2 PII Identification and Handling

PII Fields:

- Name
- Surname
- SA ID Number
- Phone Number

Handling Approach:

- SA ID and Phone Number are masked in logs
- Name and Surname are not logged
- Logs reference only CustomerID and masked values

```
Valid record. Processing Row C017 | ID: *****7082 | Phone: *****9988
Row C017 submitted to CRM successfully
Processing row: C018
Valid record. Processing Row C018 | ID: *****4085 | Phone: *****5544
Row C018 submitted to CRM successfully
Processing row: C019
Valid record. Processing Row C019 | ID: *****2087 | Phone: *****3344
Row C019 submitted to CRM successfully
Processing row: C020
Valid record. Processing Row C020 | ID: *****6083 | Phone: *****6677
Row C020 submitted to CRM successfully
```

4. Exception Handling (PA0203 & IAC0203)

Scenario A: Missing SA ID

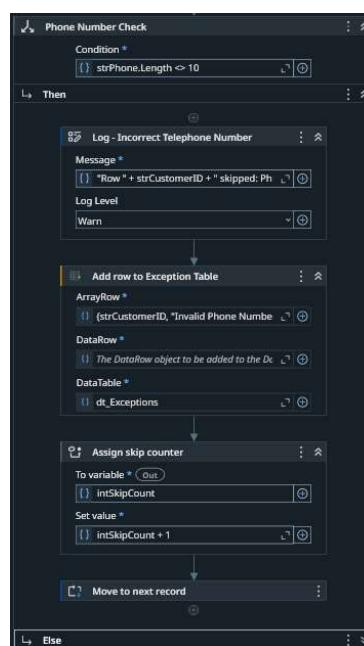
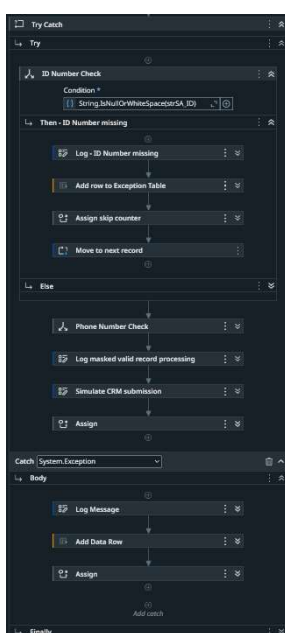
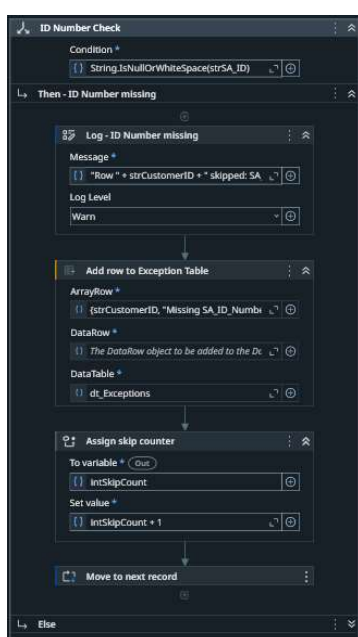
- Exception Type: Business Exception
- Handling: Log, add to exception table, increment skip counter, continue

Scenario B: CRM System Unavailable

- Exception Type: System Exception
- Handling: Try-Catch, log error, add to exception table

Scenario C: Invalid Phone Number

- Exception Type: Business Exception
- Handling: Clean with Regex, validate, log, add to exception table, skip



5. Reusability (IAC0201)

Note

This workflow was not created this section just covers what could be done.

Description:

Reusable workflow to submit validated customer data to CRM.

Input Arguments:

- Name
- Surname
- SA ID
- Phone Number

Output Arguments:

- Success Status
- Error Message

Pre-Conditions:

- CRM accessible
- Data validated

Post-Conditions:

- Customer added OR error returned

6. Solution Development (Task 5)

Implementation uses UiPath Studio with Read CSV, For Each Row, If conditions, Regex cleaning, Try-Catch, and Write CSV.

Scripting used:

```
Regex.Replace(value, "[^0-9]", "")
```

Operating System considerations include file paths and selector differences.

7. Execution Results

Final test run:

Success: 18 | Skipped: 2 | Total: 20

```
△ Row C016 skipped: SA_ID_Number is missing
○ Processing row: C017
○ Valid record. Processing Row C017 | ID: *****7082 | Phone: *****9988
○ Row C017 submitted to CRM successfully
○ Processing row: C018
○ Valid record. Processing Row C018 | ID: *****4085 | Phone: *****5544
○ Row C018 submitted to CRM successfully
○ Processing row: C019
○ Valid record. Processing Row C019 | ID: *****2087 | Phone: *****3344
○ Row C019 submitted to CRM successfully
○ Processing row: C020
○ Valid record. Processing Row C020 | ID: *****6083 | Phone: *****6677
○ Row C020 submitted to CRM successfully
○ Process completed. Success: 18 | Skipped: 2 | Total: 20
○ Exceptions report saved to Data\Exceptions\Skipped_Records.csv
○ MzansiConnect Customer Onboarding process ended at 2026-04-27 17:28:35
○ MzansiConnect_CustomerOnboarding execution ended in: 00:00:01
```

	Standard	Standard	Standard
1	CustomerID	Reason	Timestamp
2	text		
3	C012	Invalid Phone Number	2026-04-27 17:28:34
4	C016	Missing SA_ID_Number	2026-04-27 17:28:35

8. Conclusion

The automation meets all requirements including validation, exception handling, logging, and reusable design.